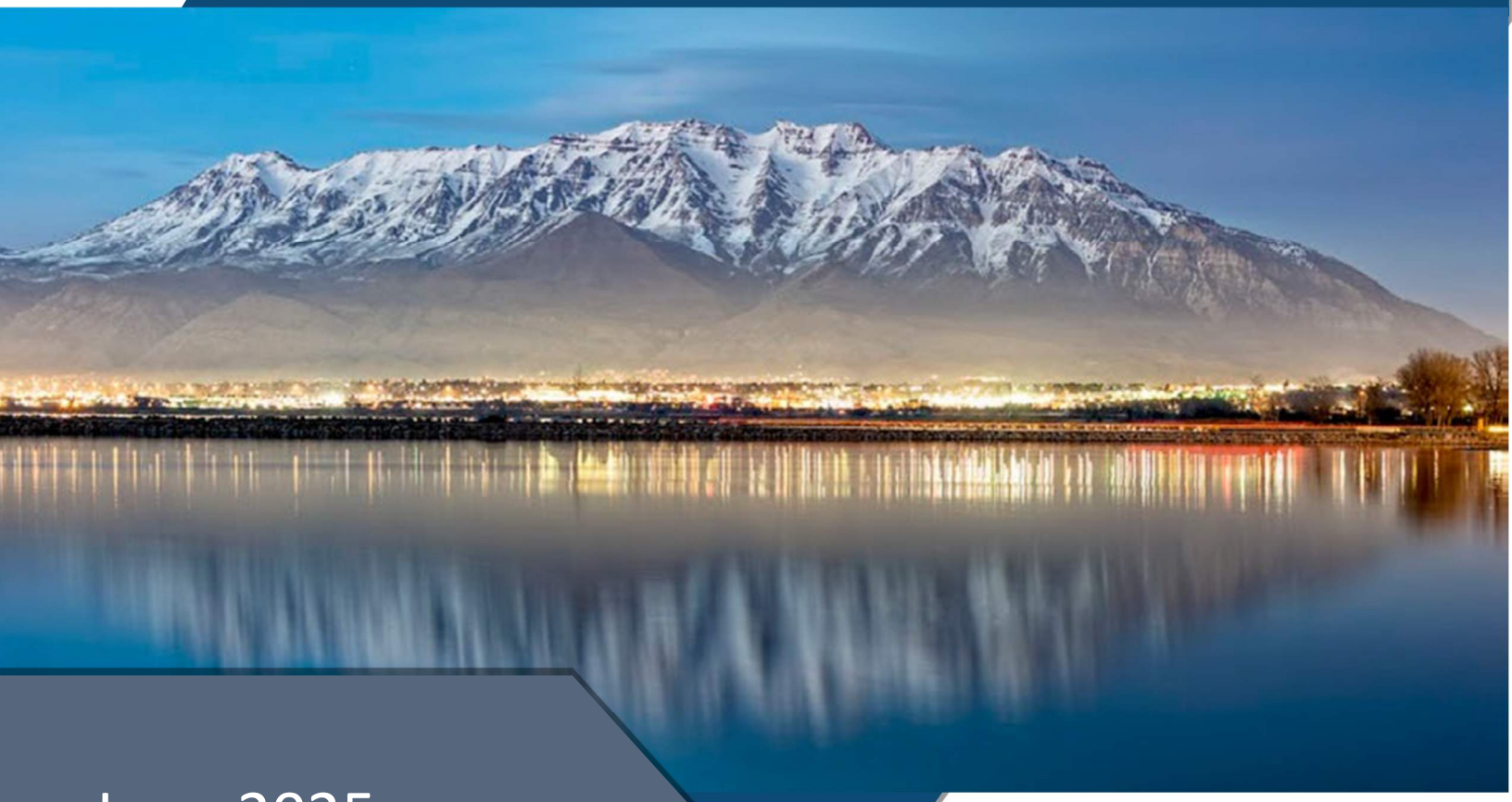
VINEYARD

Access Management Manual



June 2025

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EXECUTIVE SUMMARY

As traffic volumes and congestion continue to grow on city roads, it is important for Vineyard City engineers, planners, and decision makers to implement and enforce sound transportation policies to maximize the capital investments in transportation infrastructure. Well-managed (access-controlled) corridors can last years beyond a less restrictive access-controlled corridor as well as provide safety and economic benefits. Access management can help preserve the capital improvement dollars spent on infrastructure by preserving the capacity and prolonging the need for roadway widening projects with a well-crafted and managed plan for the various roadway cross-sections.

The purpose of this Vineyard City Access Management Manual is to provide access management recommendations to City staff and City decision makers so that appropriate and beneficial access control can be implemented throughout the city thus providing operational, safety, and economic benefits to Vineyard City. This manual has been prepared to serve as a supplementary volume to the Vineyard City Transportation Master Plan so that each document can be updated independently, as needed to keep up with the changing environment.

CONTENTS

Chapter II of this manual provides a literature review of access management principles and techniques by drawing on past research at the international, national, and state levels. Operational, safety, and economic benefits of access management are included within this review.

Chapter III provides a state-of-the-practice review of access management requirements including the Utah Department of Transportation (UDOT) access management program.

Chapter IV presents the recommended access management requirements for several access related design criteria including their location, spacing, and design characteristics. On-street parking and intersection lighting are also discussed.

Chapter V outlines the appeals process for variances to the access management standards.

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INTRODUCTION

BACKGROUND

As traffic volumes and congestion continue to grow on Vineyard City roads, it is important for city engineers, planners, and decision makers to implement and enforce sound transportation policies to maximize the capital investments in transportation infrastructure. A well-managed and access-controlled corridor can last years beyond a less restrictive corridor as well as provide safety and economic benefits. Access management can help preserve the capital improvement dollars spent on infrastructure by preserving the capacity and prolonging the need for roadway widening projects with a well-crafted and managed plan for the various roadway cross-sections.

> Access Management:
>
> "The coordinated planning, regulation, and design of access between roadways and land development."
>
> TRB Access Management Manual, 2014

The purpose of this Vineyard City Access Management Manual is to provide access management standards and recommendations to staff and decision makers so that appropriate access control can be implemented throughout the roadway network to provide operational, safety, and economic benefits to Vineyard City. This manual has been prepared to serve as a supplementary volume to the Vineyard City Transportation Master Plan (TMP).

COMPONENTS OF AN EFFECTIVE ACCESS MANAGEMENT PLAN

According to the National Cooperative Research Program (NCHRP) Report 548, A Guidebook for Including Access Management in Transportation Planning (2005), seven key components of an effective access management plan are as follows:

1. Developing and applying an access classification system that assigns access management standards to roadways in accordance with their level of importance to mobility. This classification parallels the roadway functional classification.
2. Planning, designing, and maintaining road systems based on their access classification.
3. Defining the level of access permitted to each classification (e.g., full, limited turns, traffic control type, etc.).
4. Establishing criteria for the spacing of signalized and non-signalized intersections as well as corner clearance and intersection distances from interchanges.

5. Applying agreed upon engineering standards that include appropriate geometric design criteria and traffic engineering measures for each type of access.
6. Establishing policies, regulations, and permitting procedures to implement the listed components.
7. Ensuring coordination with and supportive actions by local jurisdictions exercising their land use planning authority as well as their development permitting and reviewing authority.

This manual addresses the first five items listed above. Beyond reviewing the current UDOT access management requirements, this document does not seek to coordinate with adjacent jurisdictions regarding access management. It is recommended that this type of coordination be done as part of the master planning process that is updated from time to time by Vineyard City and other adjacent cities.

Included in Appendix A is the most recent DRAFT of the Vineyard City Transportation Master Plan Map.

ACCESS MANAGEMENT PRINCIPLES

INTRODUCTION

This chapter discusses several access management principles including access versus mobility, functional classification, access management techniques, and benefits of access management. The purpose of this chapter is to provide sufficient background regarding access management so that the reader can understand the importance of access management principles and techniques.

ACCESS VERSUS MOBILITY

The need for access management stems from the conflicting functions of any given roadway. All roads must provide some level of mobility, which means the ability to travel through a corridor, and some level of access, which means the ability to access some land use at either end of the trip. Generally speaking, the more mobility a road has, the less direct access can be provided to adjacent land uses. Alternatively, the more direct access that is provided to adjacent land use, the less mobility a given road will have.

Extreme examples of this principle are freeways and residential streets. A freeway typically has excellent mobility (high speeds and longer trips are possible) provided that demand does not exceed capacity. However, access is limited to grade separated interchanges and slip ramps only. Direct access to adjacent land use via driveways is prohibited. Conversely, a residential street has very little mobility (speeds are low and it is difficult to make longer trips). However, full access is provided to every adjacent property.

As is shown in Figure 1, arterial streets should have higher levels of mobility with lower levels of access, while local streets should have high levels of access. Collector streets have more of a balance between access and mobility providing more access than an arterial, but at the cost of having lower traveling speeds.

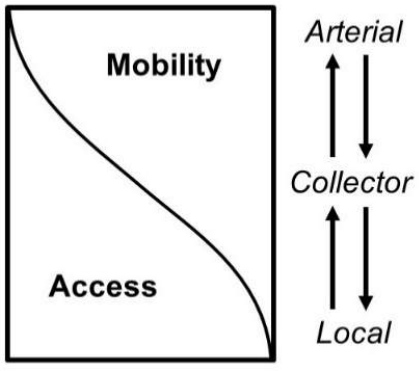


Figure 1: Access and mobility relative to functional classification

FUNCTIONAL CLASSIFICATION

Roads are broken down into a hierarchal system and given a functional classification. This is done in order to determine the purpose of a given street as well as the appropriate access classification. The higher a street classification the more mobility it provides with limited access, whereas a lower street classification has less mobility, but more access. Currently, Vineyard City has classified its roads into four classifications of public roads. Public roads include Major Arterial, Minor Arterial, Major Collector, Minor Collector, and Local Roads. These classifications are discussed in greater detail in the following sections:

ARTERIAL STREETS

Arterials are designed to have greater mobility and connect traffic between population centers and regional attractions. Because of their increased mobility, arterials typically have higher speeds and a high degree of access control. Most access to major arterials should be provided at signalized intersections and some unsignalized public streets. Private access driveways should be discouraged wherever possible. Typical cross-sections for arterial streets are shown in Figure 2.

The City's major arterials have a minimum right-of-way of 108 feet. Speed limits typically range from 30 mph to 45 mph. A major arterial street typically has two travel lanes per direction and a center two-way left-turn lane (TWLTL). The TWLTL can be replaced with a center landscaped median where left-turn movements do not need to be accommodated for access management.

The City's minor arterials have a minimum right-of-way width of 73 feet with speeds ranging from 25 mph to 35 mph. A minor arterial street typically has one travel lane in each direction separated by a center TWLTL. Little direct access is provided on minor arterials, and higher speeds allow for good mobility on these arterial streets, with slightly more access and slower speeds on minor arterials.

COLLECTOR STREETS

A collector street provides a more balanced approach towards mobility and access and provides connections between local streets and arterial streets. Collectors are often used for moderate length trips within a community. Collectors typically have two- or three-lane cross sections with parking, bike lanes, or shoulders on either side. Commercial access and local street access to collector streets is common, however, residential driveway access should be limited where possible on collectors that have higher volumes of traffic and higher speeds. Speed limits on collectors typically range from 25 to 30 mph. Typical cross-sections for collector streets are shown in Figure 3.

A major collector typically has a minimum right-of-way width of 88 feet with one travel lane in each direction, a TWLTL, sidewalks, and park strips. Some direct access is provided to adjacent properties, but reasonable speeds can also be obtained. A minor collector typically has a minimum right-of-way of 63 feet with one travel lane in each direction, no center TWLTL, sidewalks, and park strips.

LOCAL STREETS

A local street can provide full access to all adjacent land use, and usually provides little mobility. Because of this, trips on local streets are typically short and usually are between one land use and the nearest collector street. Local streets are typically wide enough for two-way travel, and many have space for on-street parking on either both, or one side of the street. The majority of local streets are residential roads which have speed limits of 25 mph. Figure 4 shows a typical cross-section of a local street with a minimum right-of-way of 58 feet.

ACCESS MANAGEMENT CONCEPTS

The following subsections briefly discuss several access management concepts and techniques currently being utilized in Utah and around the country.

TRAFFIC SIGNAL SPACING

Traffic signals should be spaced evenly and generally no closer than 1/2 mile apart. Closely or irregularly spaced signalized intersections have shown to increase delay and crash rates as good coordination is not possible. Queuing from one signalized intersection can also interfere with another closely spaced intersection. While 1/2 mile spacing has been shown to be optimal, shorter spacing (as close as 1/4 mile) may be acceptable where the demand warrants signals that close.

UNSIGNALIZED STREET AND ACCESS SPACING

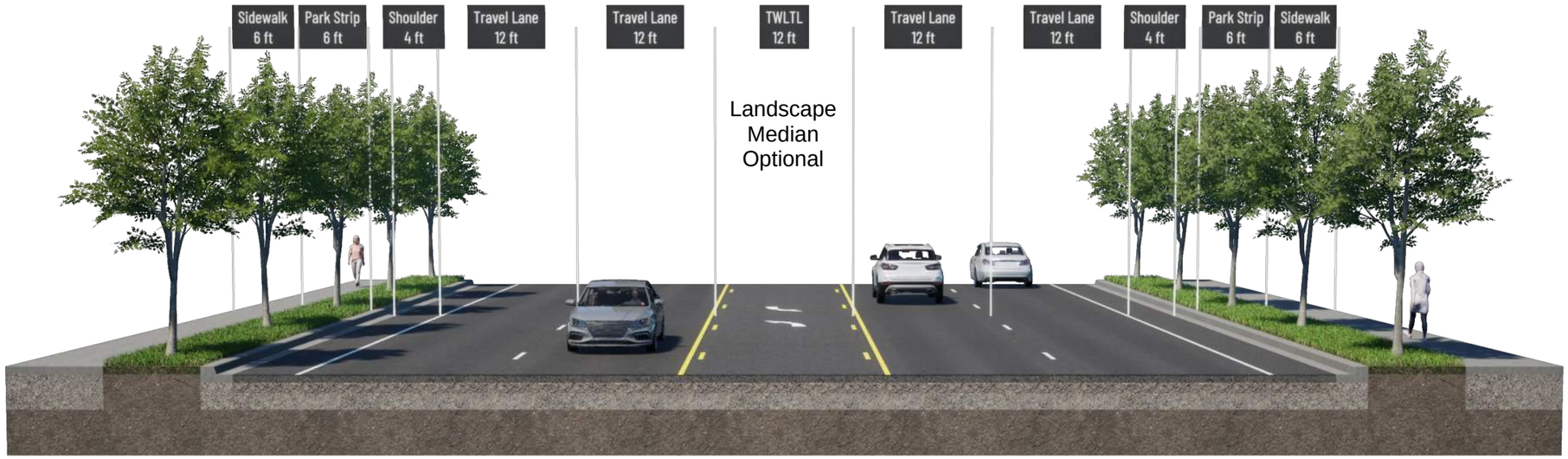
Unsignalized street spacing refers to the distance between two unsignalized intersections which can include public streets or private accesses to City roads. Research has shown that increasing the frequency of accesses (also referred to as access density) can increase the crash rate of a corridor as more conflict points are added. Average speeds can also be lowered as the additional accesses create “friction” from multiple vehicles entering and exiting the traffic stream at multiple locations.

CORNER CLEARANCE CRITERIA

The TRB Access Management Manual defines corner clearance as the distance from a public or private street to the nearest driveway (TRB 2017). The American Association of State Highway and Transportation Officials (AASHTO), *A Policy on Geometric Design of Highways and Streets*, also known as the “Green Book,” further states that driveways should not be placed within the functional area of an intersection which includes the queue storage distance, maneuver distance, and perception-reaction distance as these areas are critical areas for the safe and efficient operations of an intersection. Inadequate corner clearance can cause driveway blockages as well as decrease the overall safety of an intersection.

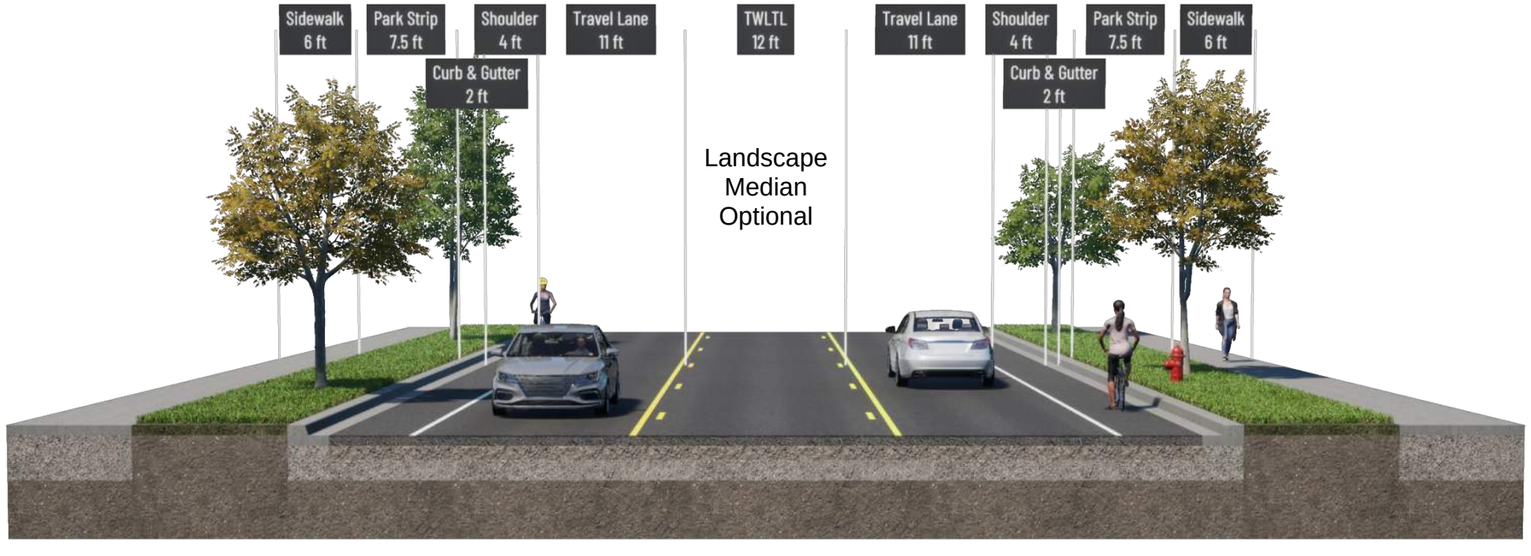
Major Arterial (5 Lanes)

Major Arterial (5 Lanes)
96+' *ROW*



Minor Arterial (3 Lanes)

Minor Arterial (3 Lanes)
73' *ROW*



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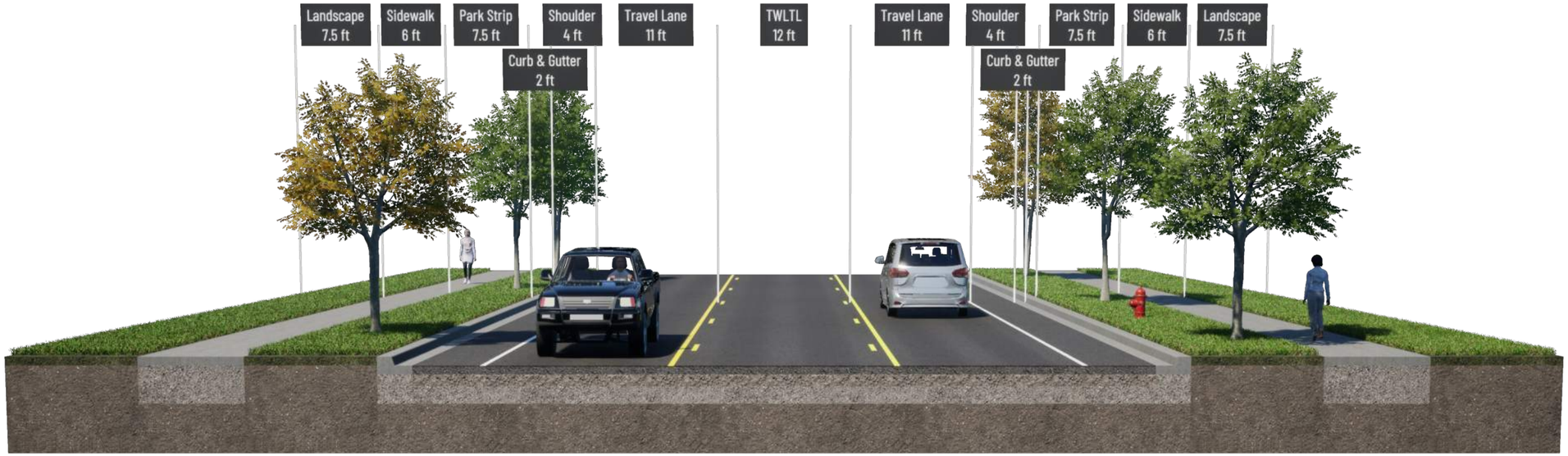
Arterial Cross Sections

Vineyard TMP

DATE 01/17/2025
PROJECT UT23-2453
Figure 2

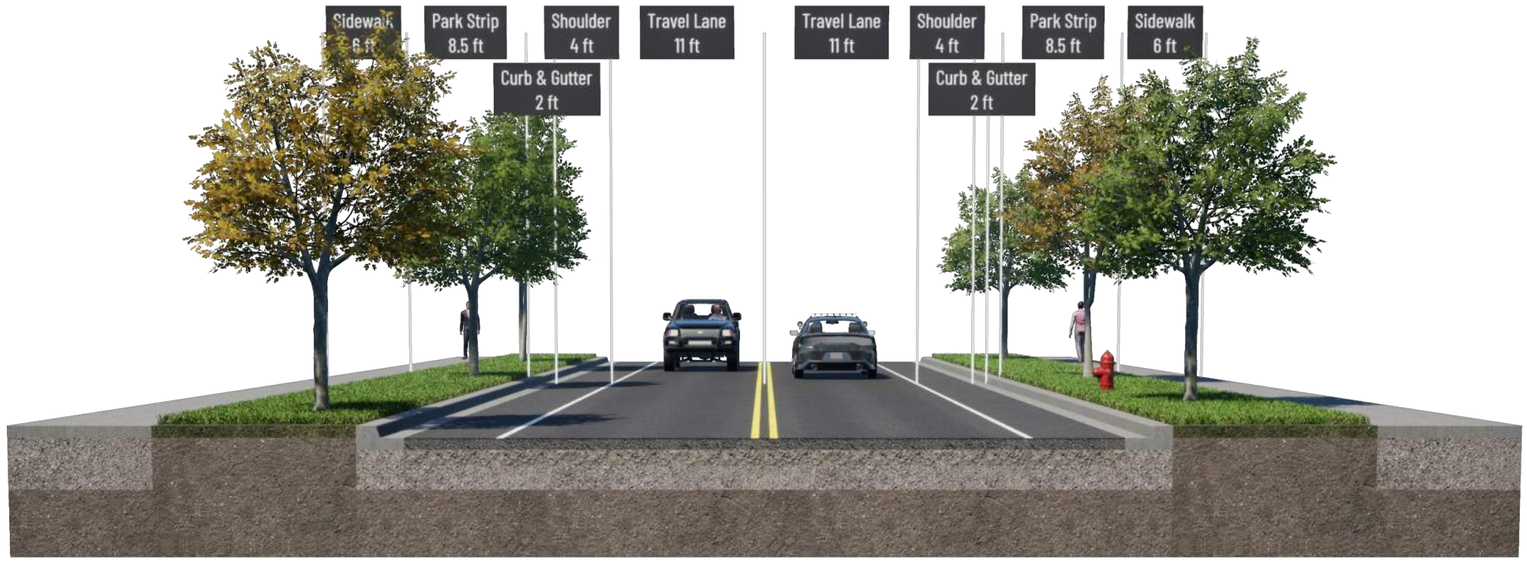
Major Collector (3 Lanes)

Major Collector (3 Lanes)
88′ *ROW*



Minor Collector (2 Lanes)

Minor Collector (2 Lanes)
63′ *ROW*



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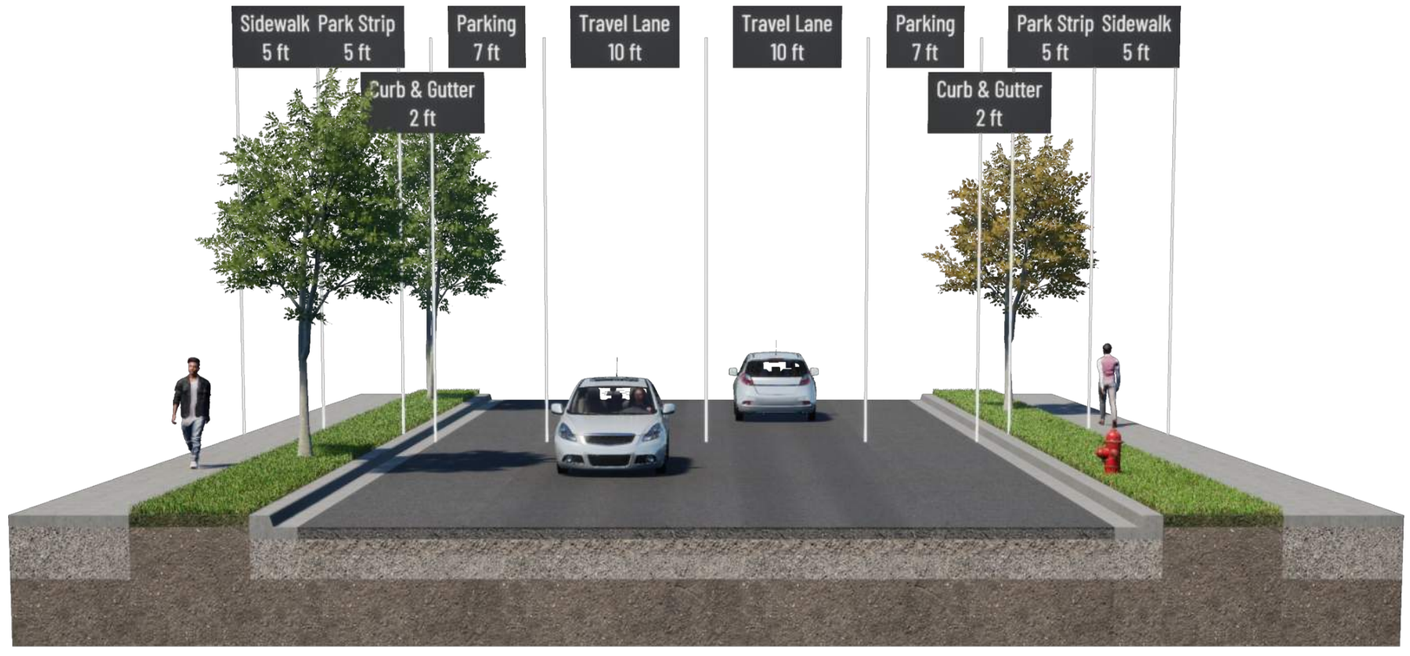
Collector Cross Sections

Vineyard TMP

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PROJECT UT23-2453
Figure 3

Local (2 Lanes)

Local (2 Lanes)
58' *ROW*



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Local Cross Section

Vineyard TMP

DATE 01/17/2025
PROJECT UT23-2453
Figure 4

MEDIAN TREATMENTS

The two most common types of medians include TWLTL areas and raised island medians. TWLTLs are safer and more efficient than no median at all; especially in commercial areas because they provide space for left-turning vehicles to exit the through traffic stream and wait for on-coming gaps to complete the left-turn movement. However, national research has shown that raised medians are even more effective in promoting safety because they provide a physical barrier to opposing directions of travel and help consolidate accesses and left-turn movements to predetermined locations. Raised medians are very effective for commercial corridors with high access density and locations with poor corner clearances and are often used as a retrofit access management technique instead of trying to remove existing accesses.

TURN LANES

Providing left-turn lanes is recommended because it removes traffic waiting for gaps out of the main traffic stream. A TWLTL or raised median accomplishes this objective on the main road. Left-turn lanes are also recommended on the minor street approaches or driveways, especially when the streets or driveways oppose another street or driveway. Opposing left-turn lanes help improve sight distance for the opposite left-turn movement, thus improving the safety of the intersection.

Exclusive right-turn lanes are also a form of access management, especially along busy corridors where there are a lot of right-turning vehicles. Because right-turn movements are usually completed at low speeds (approximately 10 mph), an exclusive right-turn lane provides refuge while the vehicle decelerates to make the right turn. At signalized intersections where right-turn-on-red movements are permitted, an exclusive right-turn lane decreases the delay for right-turning vehicles.

U-TURNS AS ALTERNATIVES TO DIRECT LEFT TURNS

U-turns as alternative to direct left turns are usually operated in conjunction with raised medians. As mid-block access is limited to right-in/right-out (RIRO) access only, left-turn ingress and egress movements must often make U-turns at other locations. These U-turns can be accommodated by having mid-block median opening with U-turns permitted, or by allowing U-turns at the next signalized intersection. When raised medians are present, some accommodation must be made for U-turns. Research has shown that U-turns as alternatives to direct left turns are safer than allowing direct left turns. Roundabouts can be an effective and safe way to make a U-turn movement, as long as it does not cause operation issues.

ACCESS SEPARATION FROM FREEWAY INTERCHANGES

Freeway interchanges have very high traffic volumes and often closely spaced signalized intersections. Research has shown that limiting access in close proximity to an interchange can reduce the need for complex signal timings, heavy weaving maneuvers, and queuing problems. Establishing good intersection spacing adjacent to freeway interchanges is crucial even if the interchange area is not very developed.

FRONTAGE/BACKAGE ROADS

Frontage roads are an effective access management alternative because they can provide increased access to property adjacent to a principal arterial or freeway with periodic access to the principal arterial at well-spaced locations. Frontage roads act like a collector street but allow good visibility to the adjacent corridor. Frontage roads are often referred to as backage roads when they are placed behind adjacent buildings, thus providing even better visibility to the corridor.

ROUNDABOUTS

Roundabouts are a form of access management because they can provide efficient and safe access to side streets as an alternative to signalized or stop-controlled intersections. While not necessarily the best solution for all intersections, roundabouts are especially effective where approach volumes are fairly balanced and where a signal or all-way stop-control would cause excessive delay most of the day. Safety benefits of roundabouts are well documented in the literature as crashes at roundabouts are much less severe than crashes at signalized intersections because they are usually rear-end or side-swipe collisions and not right-angle or head-on collisions as is often the case at signalized intersections. Additionally, crashes at roundabouts typically occur at lower speeds, thus reducing the overall severity.

SHARED (CROSS) ACCESS

In commercial corridors, an effective measure to consolidate access and limit access points on an arterial road is by providing shared access or cross access. Shared access is usually accomplished by straddling an access for two separate properties on the property line. This decreases the number of access points to the main road which improves operations and safety as discussed above as well as pedestrian safety as the number of sidewalk crossings are limited. Development and maintenance costs may also be reduced.

Cross access usually is accomplished by connecting the parking lots of two adjacent properties so that vehicles can cross between the two properties without using the public street. Cross access can be provided so that fewer accesses can be utilized for several adjacent properties, or, even if the number of accesses is not reduced, cross access can limit the number of trips on the adjacent roadway if vehicles are accessing multiple land uses. For example, trips could be reduced when an office building has cross access with a fast-food restaurant, or when a big-box retail establishment has cross access with a bank.

ACCESS ALIGNMENT/OFFSET

Good access alignment and appropriate offsets are a form of access management. According to the AASHTO Green Book, “To reduce costs and crash frequencies, intersecting roads should meet at, or nearly at, right angles, unless roundabouts are utilized. Roads intersecting at acute angles need extensive turning roadway areas and tend to limit visibility. Acute-angle intersections also increase the exposure time for the vehicles crossing the main traffic flow.” (AASHTO 2018, pg. 9-31). In addition, acute-angle

intersections are particularly difficult for elderly drivers because of the lack of flexibility in the neck of an elderly driver to check for gaps more than 90 degrees to the left or right. The AASHTO Green Book states that providing an angle of at least 75 degrees provides most of the benefits of a 90-degree intersection angle (AASHTO 2018, pg. 9-33).

Whenever possible, driveways or side streets should be aligned directly opposite from driveways or streets on the opposite side of the main road. This can increase safety as sight distance is better for opposing left turns and as it can eliminate overlapping left turns on the main road. If intersection offset must occur, it should be designed such that a vehicle waiting to turn left into one access does not occupy the same space as a vehicle waiting to turn left into the other access. Figure 5 illustrates the recommended and non-recommended types of offset.

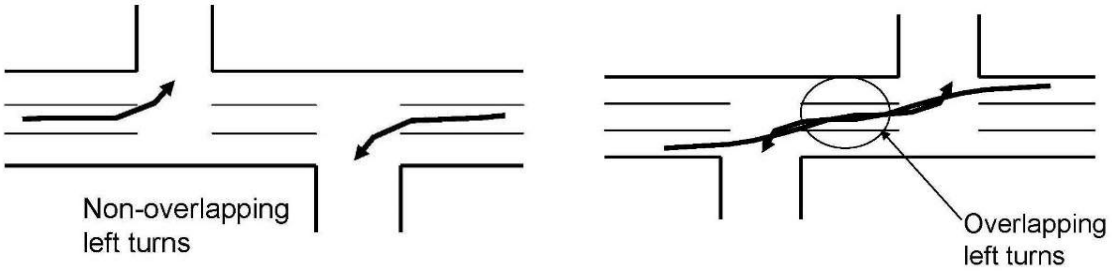


Figure 5: Offset intersection scenarios

Accesses should also be avoided near tight horizontal or vertical curves as appropriate sight distance may be difficult to obtain.

ACCESS CONTROL

Certain movements of an access can be limited or controlled to reduce conflicts thereby increasing safety and reducing delay. The most common is to designate an access as a RIRO access. A RIRO access allows only right-turn ingress and right-turn egress movements. Left turns in or out as well as crossing movements (if applicable) are not permitted. A similar, but less restrictive type of access is a three-quarter access, of which two configurations are possible. A three-quarter access can allow right ingress, right egress, and either left ingress or left egress movements, thereby eliminating only one of the four movement types (thus the name “three-quarter access”).

The effectiveness of a RIRO or three-quarter access is dependent upon the physical controls put into place for these types of accesses. Compliance is low when adequate signing and raised curbs are not present. The most effective way to enforce a limited access driveway is by use of a raised median. A raised median makes left turns physically impossible, and a directional median opening can allow either the left-ingress or left-egress movements when desired. Simply placing signs or even installing a “pork-chop” island may not prevent many drivers from making illegal turns, thus significantly reducing their effectiveness as

access management techniques. Figure 6 shows a vehicle turning left into a RIRO access. Figure 7 shows a vehicle making a left turn out of a RIRO access. Figure 8 and Figure 9 show two RIRO accesses that are operating more effectively because of medians enforcing the desired access management.



Figure 6: Vehicle making an illegal left turn into a RIRO

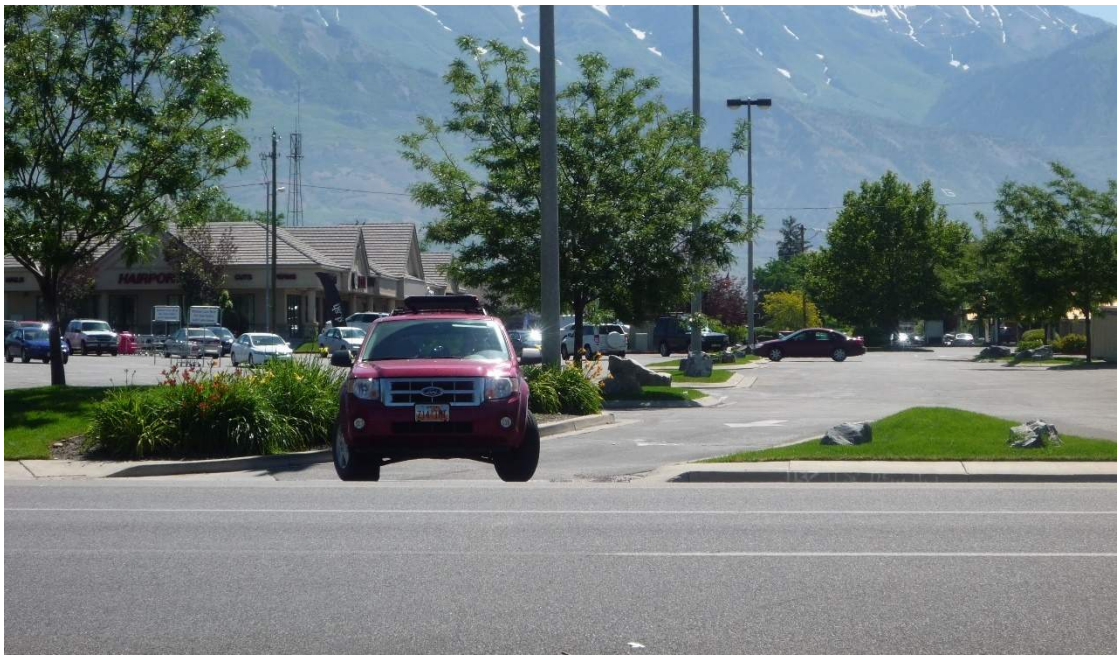


Figure 7: Vehicle making a left turn out of a RIRO

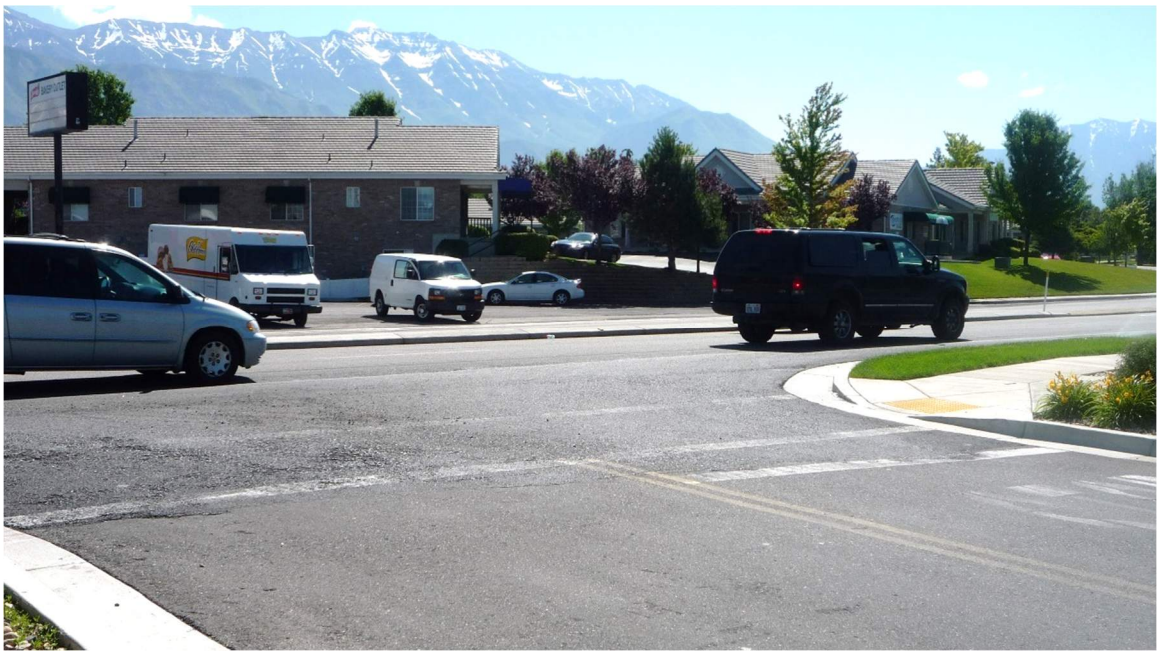


Figure 8: Raised island enforces RIRO



Figure 9: Channelized island and delineators enforce RIRO

ACCESS CHARACTERISTICS

Appropriate design of accesses to a roadway is a critical component of access management. Poorly designed accesses can create safety and operational hazards along a corridor to both through traffic and

turning traffic. The main design features of an access are the curve radii and the width of the access. Many factors influence the design including the design vehicle, design speed, traffic volume utilizing the access, and pedestrian presence.

The design vehicle used to create an access can affect the width and corner radius, especially if a driveway is designed for large trucks. A smaller corner radius can cause vehicles entering an access to cross the centerline of the access road; therefore, when heavy volumes are present, smaller radii require wider driveways. The opposite is also true because with a large enough radius, even a large vehicle will not need to cross the center line of the driveway. However, large radii increase the distance for a pedestrian to cross the access, therefore in areas with high pedestrian presence, a smaller radius may be ideal.

The type of curb opening can also affect traffic operations. The three main types of curb openings include curb returns, dropped curbs, and dustpan curbs. However, dropped curb and dustpan designs usually require much larger driveway widths as their effective radii are relatively small.

The vertical profile of an access is important because when abrupt transitions occur between the cross-slope of the roadway and the slope of the driveway, the front or rear end of a vehicle can scrape the road when a vehicle enters or exits the driveway. This causes drivers to make turns into or out of driveways at angles larger than 90 degrees which in turn can cause safety hazards, especially if the driveway is well utilized. Care should be taken to avoid an abrupt change in slopes so that the driveway can be utilized as intended without damaging vehicles. Figure 10 shows an appropriate vertical curve provided for smooth transition on this sloped access driveway.



Figure 10: Smooth vertical curve transition

The throat length of an access can also be a critical design element of an access that can impact safety and operations both on and off of the site. The driveway throat length refers to the distance of the driveway between the street and the first on-site parking area or intersection encountered. Insufficient throat length can create queuing both on- and off-site. The throat length must be long enough to handle queuing of vehicles attempting to exit the site as well as for queuing occurring as vehicles enter the site and wait for parking spots or make movements within the parking lot. Figure 11 shows an example of appropriate throat length for a driveway into a large development.



Figure 11: Example of good access throat length

INTERSECTION LIGHTING

Lighting can improve the safety and efficiency of intersections. The AASHTO Green Book (2018) recommends that lighting be present in suburban, urban, and urban core contexts "where there are concentrations of pedestrians and roadside intersectional interferences…" (pg. 3-188). Beyond traffic safety and operational benefits, street and intersection lighting can also benefit crime prevention and other community needs and should therefore be coordinated with the city agency responsible for these programs.

Intersection lighting is especially recommended when the intersection has channelized movements, large turning radii, wide cross sections, and high pedestrian volumes. Figure 12 shows typical intersection lighting.

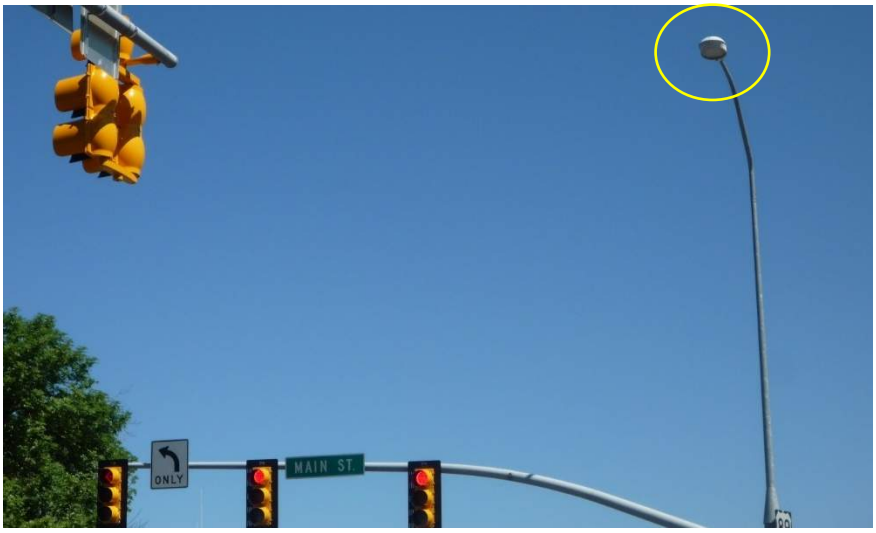


Figure 12: Typical intersection lighting

Three cities were surveyed to determine their intersection and street lighting requirements are. These requirements are summarized in Table 1 below.

Table 1: Intersection and Street Lighting Requirements

Jurisdiction	Intersection Lighting Required	Spacing Required		
		Residential Streets	Collector Street	Arterial Streets
Lehi, Utah	Yes	400' Staggered	400' Staggered	400' Staggered
Provo, Utah	Yes	Mid-Block	Per IES Guidelines[1]	Per IES Guidelines[1]
South Jordan, Utah	Yes	200'-250' Staggered	300'-350' Staggered	300'-350' Staggered

1. Illuminating Engineers Society

BENEFITS OF ACCESS MANAGEMENT

The three primary benefits of access management include operational benefits, safety benefits, and economic benefits. These three are discussed in the following sections in greater detail.

OPERATIONAL BENEFITS

Good access management can result in travel times and delays being reduced and road capacity being increased. Often this can result in a road functioning at appropriate levels of service despite increased

traffic growth. This in turn can save money for the agency responsible for the road as capacity enhancing projects such as road widening can be pushed later into the future.

SAFETY BENEFITS

Because left-turn maneuvers are typically associated with the more severe types of crashes, minimizing these conflicts through access management can improve the overall safety of a corridor. Both the quantity and severity of crashes has been shown to be reduced through implementation of access management. Sometimes less-severe crashes such as rear-end crashes increase with certain types of access management; however, the benefits of reducing right-angle crashes far outweigh the cost of additional minor crashes.

Adequate access spacing in busy urban corridors also helps reduce the decisions and workload normally required as drivers pass driveways and other intersections. This allows drivers to be more vigilant to other conditions thus improving the overall safety of the corridor.

ECONOMIC BENEFITS

A large concern of business owners and developers is that reducing direct access of businesses to major roads will hurt sales because access management techniques will make access to businesses more inconvenient, thus reducing the number of customers that will patronize the businesses. In fact, the Federal Highway Administration (FHWA) found that the opposite was true. In a study released in 2006 (Safe Access is Good for Business), the FHWA reported that before and after studies in several states showed that “the vast majority of businesses do as well or better after the access management projects are completed” because the overall travel time to access the businesses are reduced due to reductions in delay. This in turn actually makes patronizing the establishments more convenient and can increase the market area as travel times are reduced. Developments that are difficult to enter and exit due to poor access management and congestion become less attractive to customers.

ACCESS MANAGEMENT STATE-OF-THE-PRACTICE

INTRODUCTION

This chapter discusses the current state-of-the-practice for access management related standards including the current policies of Utah Department of Transportation (UDOT) and several other cities in Utah.

UTAH DEPARTMENT OF TRANSPORTATION (UDOT)

UDOT began a formal access management program in 2003 with Administrative Rule R930-6, *Accommodation of Utilities and the Control and Protection of State Highway Rights of Way*. Chapter 7 of this rule outlines the access management plan for all UDOT-owned roads. Several topics are coved by this program including:

- Access management classifications (Category 1 through 9)
- Access spacing requirements for each classification including signal spacing, street spacing, access spacing, and spacing from crossroads to freeway interchanges.
- Access design standards and specifications
- Turn lane requirements
- Permitting and variance processes

UDOT ACCESS LOCATION AND DESIGN STANDARDS

Table 2 shows the access spacing requirements for UDOT roads. UDOT's access spacing requirements serve as a good starting point for Vineyard access spacing requirements, especially for Vineyard's higher classification roadways. However, UDOT's requirements do not cover lower classified roads such as residential streets and collector streets.

Table 3 shows recommended minimum and maximum access widths for driveways on UDOT facilities based on land use type. Table 4 shows the recommended minimum and maximum radii for urban accesses on UDOT facilities. Table 5 shows the desirable and minimum approach angles of access to UDOT facilities, that is, the maximum allowable skew of an access.

Table 2: UDOT State Highway Access Management Standards

	Category	Minimum Signal Spacing (ft)	Minimum Street Spacing (ft)	Minimum Access Spacing (ft)	Minimum Interchange to Crossroad Access Spacing (ft)		
					To 1st RIRO[1]	To 1st RIRO Intersection[2]	From Last RIRO[3]
1	Freeway/Interstate	Not Allowed			Not Applicable		
2	System Priority-Rural	5,280	1,000	1,320	1,320	1,320	1,320
3	System Priority-Urban	2,640	No Unsignalized Access Permitted		1,320	1,320	1,320
4	Regional-Rural	2,640	660	500	660	1,320	500
5	Regional Priority-Urban	2,640	660	350	660	1,320	500
6	Regional-Urban	1,320	350	200	500	1,320	500
7	Community-Rural	1,320	300	150	Not Applicable		
8	Community-Urban	1,320	300	150			
9	Other	1,320	300	150			
10	Freeway One-way Frontage Road	1,320	660	Not Allowed			

1. The distance from interchange off-ramp gore area (point of widening) to the first RIRO driveway intersection.
2. The distance from the interchange off-ramp gore area (point of widening) to the first major intersection.
3. The distance from the last RIRO driveway intersection to the interchange off-ramp gore area (point of widening).

Source: UDOT R930-6-7, Table 1 (2019)

Table 3: UDOT Access Width Guidance

Land Use	Direction Use	Minimum Access Width (ft)	Maximum Access Width (ft)
Commercial or Industrial	Two-way	25	50
	One-way	16	30
Residential	Two-way or one-way	16	30
Farmland	Two-way or one-way	16	32

1. If two one-way approaches are adjacent and separated by a non-traversable median between 4 and 25 feet wide, the access is treated as one access.

Source: UDOT R930-6-7.4 (2019)

Table 4: UDOT Access Radii (Urban Areas)

Land Use	Location	Minimum Radius (ft)	Maximum Radius (ft)
Commercial, Industrial, Agricultural	Urban	30	60
	Rural	20	60
Residential	Urban	10	15
	Rural	20	30
1. Examine the vehicle profile utilizing the subject access and design appropriate radii. Source: UDOT R930-6-7.4 (2019)			

Table 5: UDOT Access Driveway Angles

Direction Use	Desirable (degrees)	Minimum (degrees)
Two-way	90	80
One-way (right-in or right-out)	90	60
Source: UDOT R930-6-7.4 (2019)		

ACCESS MANAGEMENT STANDARDS

INTRODUCTION

The following sections discuss access management standards for Vineyard City. These are based on national and local research, as well as the current state-of-the-practice by other jurisdictions as previously discussed.

ACCESS PROVISIONS

This section discusses the types of accesses that should be permitted on the various types of roadways within the City. The roadway network and land use are interconnected. Therefore, allowing certain types of land use and accesses adjacent to certain types of roadways will lead to maximizing the life and traffic operations of the roadway. Table 6 shows recommended allowable accesses for each roadway classification.

Table 6: Access Provisions

Street Classification	Commercial Access	Multi-Family Residential Access	Residential Access	Public Street Access
Arterial	Yes[1]	No[3]	No	Yes
Collector	Yes	Yes	Yes[4,5]	Yes
Local Street	No[2]	Yes	Yes	Yes

1. Preferably, commercial access to an arterial should be consolidated for several parcels and gained using a traffic signal or a public street. Where direct access is given to an arterial, if possible, the access should be a RIRO access.
2. Exceptions to this recommendation would be for small commercial developments or for secondary access to a commercial development.
3. Very large multi-family developments may have private accesses that resemble public streets in form and function, and therefore would be an exception to this recommendation.
4. Direct access and frontage of residential properties to a minor collector may result in less-than-desirable livability for residents and should be avoided where practical.
5. Direct access and frontage of residential properties to a major collector may result in less-than-desirable livability for residents and should be avoided where practical; however, if base residential zoning allows access, then a traffic impact study showing acceptable operations and safety can be mitigated, then an engineering review will determine the applicability of this standard.

The standards listed in Table 6 may not always be feasible depending on whether or not other access exists for a given parcel of land. The feasibility of these recommendations is also highly dependent upon land use and transportation planning by Vineyard City.

INTERSECTION AND ACCESS SPACING

Standards for minimum signalized, public street and private access spacing have been compiled and are shown in Table 7. Guidance for access widths and angles are summarized in Table 8, and Table 9, respectively.

Table 7: Vineyard Access Management Standards

Functional Classification	Signal Spacing[1] (ft)	Street Spacing[1] (ft)	Access Spacing[1] (ft)		
			Full	3/4[2]	RIRO[3]
Major Arterial	2,640	1,320	1,320	990	660
Minor Arterial	1,320	660	660	330	330
Major Collector	1,320	330	330		
Minor Collector	1,320	330	150		
Local	N/A	150	35		

1. Includes intersecting streets and driveways. Signal and access spacing may be adjusted by the City Engineer based on traffic warrants as found in the latest Manual on Uniform Traffic Control Devices (MUTCD) and engineering judgement.
2. Restricts left-turns and through movements from the minor approach OR left-turns from the major approach
3. Limits access to right turns into and out of the minor approach only

Table 8: Vineyard Access Width Standards

Land Use	Desired Access Width (ft)	Maximum Access Width (ft) [3]
Private Residences	Match width of driveway	30[1]
Residential Development	12 feet for each entry/exit lane	30
Commercial Development	14 feet for the entry lane & 12 feet for each exit lane	30[2]
Industrial Development	16 feet for the entry lane & 12 feet for each exit lane	40

1. For residential lots with a width less than 60 feet, the driveway width may not exceed fifty percent (50%) of the lot width.
2. Subject to the approval of the City Engineer, the maximum width may be increased to 40 feet in the C-1 General Commercial District based on traffic needs. Other Commercial Districts may be allowed a maximum width driveway of 40 feet, based on a Traffic Impact Analysis and subject to approval of the City Engineer.
3. Maximum width shall be measured at the right-of-way line.

Table 9: Vineyard Access Driveway Angles

Land Use	Direction Use	Driveway Angles	
		Desirable (degrees)	Minimum (degrees)
Commercial or Industrial	Two-way	90	75
	One-way (right-in or right-out)		60
Residential or Farm Land	Two-way or one-way		75

1. If two one-way approaches are adjacent and separated by a non-traversable median between 4 and 25 feet wide, the access is treated as one access.

CORNER CLEARANCE REQUIREMENTS

The corner clearance requirements are the same as the minimum access spacing requirements as were shown previously in Table 7. In general, minimum corner clearance can be shorter on the downstream leg of the minor street at an intersection, especially if the access is a RIRO access. This distance can be as low as 120 feet. For all other corner clearance distances, the values in Table 7 for minimum access spacing apply.

CROSS ACCESS (SHARED ACCESS) REQUIREMENTS

Access to private property from the public transportation network cannot be prohibited. However, when parcels have small frontages, especially along arterials and collectors, it is highly recommended that a cross access and/or shared access be utilized where practical. Where adjoining parcels have the same ownership, Vineyard City should require the use of cross or shared access in order to maintain the recommended access spacing requirements as previously discussed. When adjacent parcels do not have common ownership, incentives should be provided to promote shared and cross access.

INTERSECTION ALIGNMENT / OFFSETS

Where possible, accesses on opposite sides of the street should be lined up directly across from each other. Sometimes this is not possible due to site restrictions or property lines. If accesses cannot be lined up across from one another, the preferable offset is so that left turns entering each respective access do not overlap, as was discussed previously in this document and shown in Figure 5. Table 10 shows the minimum offset for intersections on opposite sides of the street when the potential for overlap does occur. However, a traffic impact study could determine that a larger offset is required depending on the level of queuing that could occur at the intersections.

Table 10: Minimum Access Offset Requirements

Street Classification	Minimum Offset (feet)[1]
Minor Arterial[2]	300
Collector	180

1. These values are a minimum requirement and are subject to change based on the results of a traffic impact study. Offset requirements are not applicable in the case of RIRO accesses such as when a raised median is present.
2. Access should only be provided to arterial streets when reasonable alternative access does not exist.

ON-STREET PARKING

Permitting on-street parking can be a tool to make a corridor more pedestrian friendly as parked vehicles provide a buffer between through traffic and pedestrians on the sidewalks. Vehicles parking and un-parking along a roadway may cause side friction that can result in through traffic being slowed down or impeded. To determine whether or not a road should have permitted on-street parking depends primarily on what the function of the roadway is. As was discussed previously, the purpose of most arterial type roads is to provide mobility, not access, whereas the purpose of a local street is to provide access and not mobility. However, another important factor is adjacent land use.

A roadway in a core business area, even if classified as an arterial, can benefit from having on-street parking, where a lower classified road, such as a minor collector within an industrial area, may not need on-street parking, therefore the cost to build and maintain this type of road will be less expensive by prohibiting on-street parking. Finally, the available cross-section width should also be considered when determining whether on-street parking should be permitted. A minimum of 7 feet of cross-section width should be provided for on-street parking on one side or 14 feet when parking is desired on both sides of the road. Typically, the gutter pan can be used for additional width for parking beyond the 7 feet of pavement.

As shown previously in Figure 2 through 4, the typical cross-sections only provide for on-street parking on local roads by default. However, if the City and/or development groups desire on-street parking along a corridor based on the land use context, on-street parking may be accommodated within the right-of-way by reducing the size of the park strip and widening the shoulder. This will also bring the sidewalk closer to the on-street parking, making it easier for users to get in and out of parked vehicles.

ACCESS CHARACTERISTICS

Standards for access characteristics discussed in this section include driveway width, curb return radii, vertical profile of driveway, driveway throat length, and minimum separation of gate on private access from public street.

All recommendations in this section assume the following:

- The design vehicle for residential and commercial accesses is a passenger car
- The design vehicle for industrial accesses is a WB-50

Although some larger vehicles such as delivery vehicles do need access to commercial areas, these types of trips do not typically occur during the peak hours of the day. The entire access (ingress and egress side) can therefore be used by a larger vehicle during off-peak times. This is more efficient than utilizing more area to make larger radii and access widths and is also more ideal for pedestrians. Access driveways that will be heavily utilized by trucks and buses during peak periods should be designed to the industrial recommendations. Table 11 shows curb radii for various intersection types depending on the functional classification of the adjacent roadway.

Table 11: Curb Return Radii

Access Type	Curb Return Radii in Feet		
	Arterial	Collector	Local
Residential (SFDU) 1 dwelling unit	n/a	n/a	n/a
Multi-Family (less than 3 units)	n/a	n/a	n/a
Multi Family (more than 3 units)	20	20	25
Commercial (1 exit lane)	20	20	25
Commercial (2 exit lanes)	20	n/a	n/a
Industrial	45	45	n/a

Source: *AASHTO Green Book* (2018)

Table 12 shows the required curb radius measured from top back of curb (TBC) for the various street intersections within Vineyard City. Deviation from these standards requires approval in writing from the City Engineer.

Table 12: City Street Curb Radii

Minor Street Classification	Curb Radii (ft) by Major Street Classification			
	Local	Collector	Minor Arterial	Major Arterial
Local	15	15	15	25
Collector	n/a	25	30	30
Minor Arterial	n/a	n/a	30	30
Major Arterial	n/a	n/a	n/a	40

Source: *AASHTO Green Book* (2018)

Table 13 contains maximum change in grade between the cross-slope of the roadway and the slope of the driveway. The maximum change in grade reported is the algebraic difference in slope. For example, if a negative 2 percent cross-slope connects to a positive 2 percent driveway slope, the algebraic change in grade is 4 percent. Likewise, if the cross-slope is negative 2 percent and the driveway slope is negative 6 percent, the algebraic change in grade is also 4 percent. Table 13 also shows the maximum change in grade with vertical curves provided.

Table 13: Change in Driveway Vertical Profile

Access Type	Maximum Change in Grade – No Vertical Curve Required[1]			Maximum Change In Grade – Vertical Curve Required		
	Arterial	Collector	Local	Arterial	Collector	Local
Residential (SFDU)	n/a	8%	10%	n/a	10%	12%
Multi-family (3 or less units)	n/a	8%	10%	n/a	10%	12%
Multi-family (4 or more units)	4%	8%	10%	5%	10%	12%
Commercial (1 exit lane)	4%	8%	10%	6%	10%	12%
Commercial (2 exit lanes)	4%	n/a	n/a	5%	n/a	n/a
Industrial	4%	-	n/a	5%	-	n/a

1. Rounding by taking care to avoid abrupt changes in grade.

Source: Transportation and Land Development (ITE, 2002)

Table 14 shows the recommended driveway throat lengths for various access types and based on what type of conflict is closest to the adjacent road. When an internal intersection is the first conflict encountered when entering a driveway, a shorter throat is typically needed then for a site where the driver will encounter parking spots after entering the access.

Table 14: Recommended Driveway Throat Length (feet)

Access Type	First conflict within the site is:	
	Parking	Internal Intersection
Residential (SFDU)	n/a	n/a
Multi-Family (3 or less units)	n/a	n/a
Multi Family (4 or more units)	75	30
Commercial (1 exit lane)	75	30
Commercial (2 exit lanes)[1]	75	50
Industrial	75	75

1. For signalized accesses with three egress lanes, throat length should be at least 200 feet long. For signalized accesses with four egress lanes, throat length should be at least 300 feet long. These lengths should also be verified by a traffic study.

Source: *Transportation and Land Development* (ITE, 2002)

Table 15 shows the minimum separation of a gate on a private access from the edge of traveled way for various design vehicles. If a sidewalk, trail, and/or bike lane exist, the edge of traveled way should be considered as the closest edge of traveled way for any of these modes of transportation (e.g. setback should be measured from the gate to the edge of sidewalk if present). This separation is necessary so that a vehicle can park in front of the gate while it is still closed without the rear of the vehicle obstructing the flow of traffic. Where the possibility of queuing exists, such as at a gated residential community, a traffic study should be completed to determine the 95th percentile queue length during the peak hour.

Table 15: Minimum Setback of a Gate on a Private Access from an Adjacent Street

Design Vehicle	Minimum Distance (feet)[1]
Passenger Car	25
Single Unit Truck	35
Bus	50
Recreational Vehicle	35
Recreational Vehicle Pulling a Trailer	60
WB-40	50
WB-62	75
WB-67	80

1. If multiple vehicles are expected to queue up at the gate, a traffic study should be completed to determine the required distance.

Source: AASHTO Green Book (2018)

INTERSECTION SIGHT DISTANCE

Appropriate intersection sight distance (ISD) should be provided for at each access and intersection based on the type of control (signal, stop sign, yield sign, roundabout, etc.), the design speed of the roadways, the design vehicle, the permitted movements at the intersection, and the cross-sectional width of the road to be traversed. Pedestrian usage can also be a factor in required intersection sight distances. The AASHTO *A Policy on Geometric Design of Highways and Streets* ("Green Book") provides standards for calculating the appropriate intersection sight distances. Figure 13 and Figure 14 show examples of poor intersection sight distance because of bushes, trees, and other structures.

This section discusses recommended intersection sight distance for several intersection types based on AASHTO Green Book (2018) recommendations. These recommendations are made for various intersection control scenarios as well as functional classifications of the intersecting roads. Additionally, sight distances to accommodate trucks have also been calculated. Scenarios discussed in the following sections include intersection sight distance for stop-controlled, yield-controlled, and uncontrolled

intersections. Sight distance requirements for a left turn from the major street onto a minor street are also given. A licensed professional engineer should calculate all intersection sight distances.



Figure 13: Sight distance obscured by landscaping features



Figure 14: Sight distance obscured by curvature and structure

For all of the scenarios, the following assumptions have been made:

- Design Speeds:
  - Access: 15 mph
  - Local: 30 mph
  - Minor Collector: 35 mph
  - Major Collector: 40 mph
  - Minor Arterial: 45 mph
  - Major Arterial: 50 mph
- Intersections are flat (all slopes are assumed 3 percent or less)
- All roads are assumed to be 2-lane facilities. Adjustments may be needed for wider roads depending on the ISD case.

All calculations are based on *AASHTO Green Book* (2018) assumptions. Because the grade of the major or minor street does affect sight distance, the values presented in the sections below should be adjusted when grades are more than 3 percent. These adjustments should be made on an individual case-by-case basis by consulting the *AASHTO Green Book*. Additionally, alternative design speeds or cross sections will also require adjustments to these sight distances.

Once intersection sight distances have been determined for a given intersection, a "clear sight triangle" can be formed which should remain clear of anything that will obstruct the driver from seeing other vehicles within the intersection sight distance. The *AASHTO Green Book* states that the height of a driver's eye in a passenger vehicle is approximately 3.5 feet. The height corresponding to the driver of a combination or single unit truck is approximately 7.6 feet. Therefore, objects should not be placed in the clear sight triangle between 3.5 and 7.6 feet in height.

Objects such as buildings, fences, and parked cars are examples of objects that will block the intersection sight distance. Landscaping features such as shrubbery and decorative rocks are acceptable as long as they are less than 3.5 feet high. Trees are acceptable as long as the trunk is located outside of the clear sight triangle and the lowest branches are at least 7.6 feet tall. The *AASHTO, Roadside Design Guide,* provides additional guidance and should be consulted for concurrence or variance.

STOP-CONTROLLED INTERSECTIONS

AASHTO defines three different cases for stop-controlled intersections:

- Case B1: Left turns from the minor road
- Case B2: Right turns from the major road
- Case B3: Crossing the major road from a minor-road approach

Intersection sight distances for stop-controlled intersections include both full-movement and RIRO intersections. Cases B1 and B3 apply to full-movement intersections and Case B2 applies to RIRO intersections.

Sight distances for each case are unique based on the design speed of the major road. When drawing the sight triangle for stop-controlled intersections, the vertex (decision point) of the sight triangle on the minor street leg should be located 14.5 feet behind the edge of the major-road traveled way. The edge of traveled way is defined as the outside striping of the closest major-road lane or the edge of pavement if there is not striping.

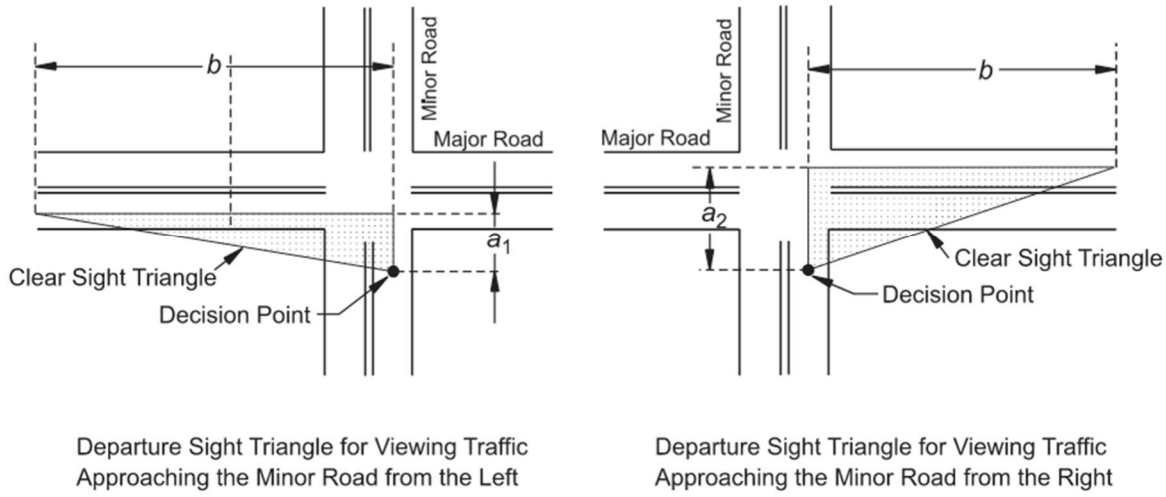
Table 16 shows the sight distance required for left-turn movements from the minor road (Case B1). Table 17 shows the sight distance required for right-turn movements from the minor road (Case B2) and crossing maneuvers from the minor road (Case B3). The value shown in the table is how far to the left and/or right for which the stopped vehicle should have a clear sight triangle from the decision point. Figure 15 illustrates these distances and the sight triangle.

Table 16: ISD Case B1 – Left-turn from Minor Road

Design Vehicle	ISD (ft) Major Street Design Speed		
	30 mph	40 mph	50 mph
Passenger Car	335	445	555
Single Unit Truck	420	560	700
Combination Truck	510	680	850
Source: *AASHTO Green Book* (2018) Case B1 (pp. 9-43 to 9-47)			

Table 17: ISD Cases B2 and B3 – Right-turn and Crossing Maneuvers from Minor Road

Design Vehicle	ISD (ft) Major Street Design Speed		
	30 mph	40 mph	50 mph
Passenger Car	290	385	480
Single Unit Truck	375	500	625
Combination Truck	465	620	775
Source: *AASHTO Green Book* (2018) Cases B2 and B3 (pp. 9-47 to 9-50)			



Departure Sight Triangle for Viewing Traffic Approaching the Minor Road from the Left

Departure Sight Triangle for Viewing Traffic Approaching the Minor Road from the Right

Figure 15: Intersection sight triangles for stop control

YIELD-CONTROLLED INTERSECTIONS

Intersection sight distance for yield-controlled intersections include two cases:

- Case C1: Crossing maneuver from the minor road
- Case C2: Left- and right-turn maneuvers from the minor road.

In both scenarios, the major street does not stop. For four-leg intersections, Cases C1 and C2 both need to be evaluated since crossing maneuvers are possible. For three-leg intersections, only Case C2 needs to be evaluated.

Table 18 shows the sight distance required for Case C1. Table 19 shows the sight distance required for Case C2. Figure 16 illustrates these distances and the sight triangles. As shown, the length of the sight triangles listed in the tables are measured to the center of the lane.

Sight distance values are only shown for passenger cars as it is assumed that roads with significant truck traffic will not be yield-controlled intersections. Also, it was assumed that the major streets do not have classifications higher than minor collector, as these types of high classification streets should at least have stop control on the minor street. If conditions are different than the assumptions stated, unique sight distance values should be calculated based on the AASHTO Green Book (2018). If an intersection location does not meet the required ISD for cases C1 and C2, stop control is required.

Table 18: ISD Case C1 – Crossing Maneuvers from Yield Control

Minor Street Design Speed	ISD (ft) by Major Street Design Speed		
	30 mph	40 mph	50 mph
20-50 mph	290	385	480
Source: *AASHTO Green Book* (2018) Case C1 (pp. 9-51 to 9-54)			

Table 19: ISD Case C2 – Left- and Right-Turn Maneuver from Yield Control

Design Vehicle	ISD (ft) by Major Street Design Speed		
	30 mph	40 mph	50 mph
Passenger Car	355	475	590
Single Unit Truck	445	590	735
Combination Truck	530	710	885
Source: *AASHTO Green Book* (2018) Case C2 (pp. 9-53 to 9-55)			

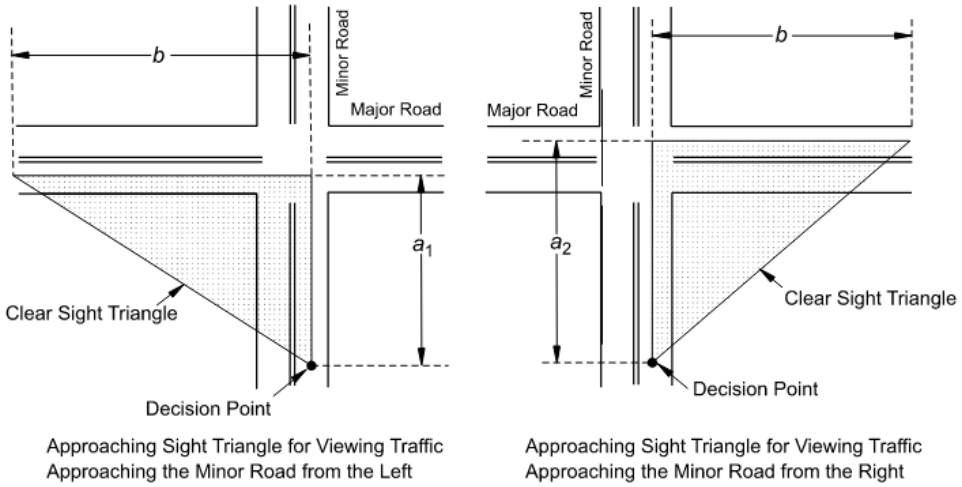


Figure 16: Intersection sight triangles for yield control

UNCONTROLLED INTERSECTIONS

Uncontrolled intersection sight distances are for those intersections where neither the minor nor the major street has any control. These streets are typically very low volume roads primarily in residential subdivisions. An uncontrolled intersection is defined as Case A by AASHTO. Table 20 shows the sight distance required at uncontrolled intersections. Only access roads and local streets were shown as any other higher classification road should have some intersection control. The distance shown in the table

corresponds to the leg of the sight distance triangle for the respective type of road. Figure 16 above illustrates the sight distance triangle for this type of intersection. Sight distance values were only calculated for passenger cars as it is assumed that roads with significant truck traffic will not be yield-controlled intersections. If an intersection location does not meet the required ISD listed in Table 20, stop or yield control should be considered.

Table 20: ISD Case A – Uncontrolled Intersection (feet)

Length of Approach Leg by Major Street Design Speed		
30 mph	40 mph	50 mph
140	195	245
Source: AASHTO Green Book (2018) Case A (pp. 9-40 to 9-41)		

LEFT TURNS FROM THE MAJOR ROAD

Intersection sight distance for left turns from the major road include any left turn movement where the driver must yield to oncoming traffic (through and right-turning vehicles) before turning left onto the minor street. This is defined as Case F by AASHTO. Table 21 shows the values for the sight distance triangles based as a function of design vehicle. Figure 17 illustrates the sight distance triangles for this type of movement.

Table 21: ISD Case F – Left Turns from Major Street

Design Vehicle	ISD (ft) by Major Street Design Speed*		
	30 mph	40 mph	50 mph
Passenger Car	245	325	405
Single Unit Truck	290	385	480
Combination Truck	335	445	555
Source: AASHTO Green Book (2018) Case F (pp. 9-56 to 9-58) **These values assume one opposing through lane. Adjustments are necessary if more than one opposing through lane or opposing right-turn lane with a shared receiving lane exist.*			

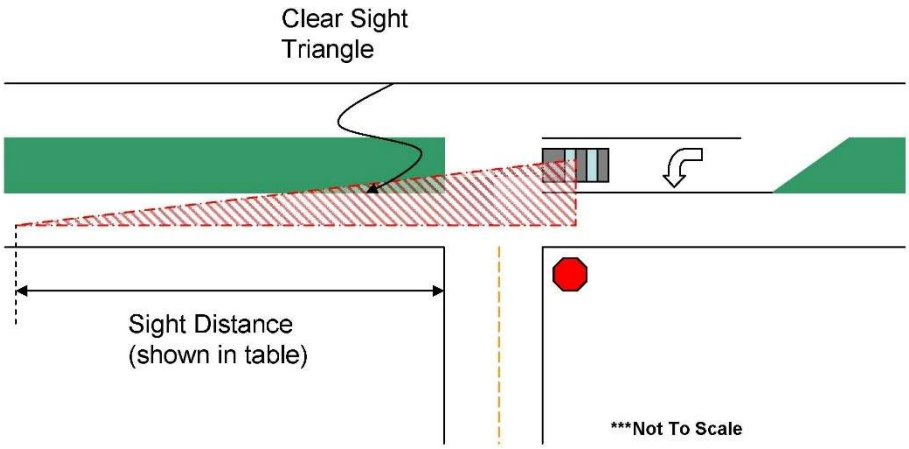


Figure 17: Intersection sight triangle for a major street left turn

VARIANCE PROCESS

An access variance grants permission to depart from the standards and requirements of the Vineyard City access management manual.

Applicants seeking a variance from the Vineyard City access management manual will submit the written request as an attachment to the formal project submittal and application. A subsequent request for a variance may be allowed in accordance with the completeness check review procedure as a supplement to a previously submitted application if Vineyard City determines that it is in the public interest to do so. The City Public Works Director or Engineer will review all requests for variance. If a variance request affects a federal aid roadway (both state highway and local roads) UDOT may need to be consulted as well.

The request for variance will specify in writing why the variance is appropriate and necessary and should be completed and stamped by a licensed professional traffic engineer. Included in the request, documentation of conditions with and without the variance with the documentation showing that the applicant has considered all practical and reasonable mitigation alternatives. The variance request results from the application of the standard or requirement of this access management manual suffered directly and solely by the applicant and is not self-created or self-imposed such as by the applicant acting with or without knowledge of the applicable standard or requirement. A variance will not be granted for procedural requirements.

The applicant is responsible for showing that the variance request meets minimum acceptable engineering, operation, and safety standards, the variance is not detrimental to the public health, welfare and safety, and the variance is reasonably necessary for the convenience and welfare of the public.

The City Public Works Director or Engineer will consider the following factors in determining that the granting of a variance will not negatively impact the current and proposed operation of the roadway:

1. The applicant has considered all other feasible alternatives to provide reasonable access to the land use or development and can demonstrate that better alternatives in terms of roadway operations are not feasible or do not exist.
2. The applicant has considered access through a shared use or cross access driveway or access point with an adjacent land use and such a shared use access or cross access is not feasible. A letter from the adjacent land owner stating that cross access or shared access is not feasible shall be provided.
3. The applicant is providing on-site or off-site traffic improvements that might offset the negative impacts of granting an access that does not meet standards.
4. The applicant has considered and demonstrated trip reduction strategies that allow the access to properly function without creating a negative impact to the roadway.

5. The applicant has provided traffic engineering or other studies to determine that the access will not degrade the efficient flow of traffic on the roadway in terms of safety, capacity, travel speed and other functional features of the roadway.

The review and final action of whether to approve or disapprove the variance will be completed within twenty (20) working days of the date of acceptance of the request for variance application.

Include the documentation of reasons for approving/denying the variance in the project files and records. State in terms and conditions of the approved permit and variance that the permittee may be required to improve, modify, eliminate, or correct the condition responsible for the variance when it is evident that the justification for the variance is no longer valid.

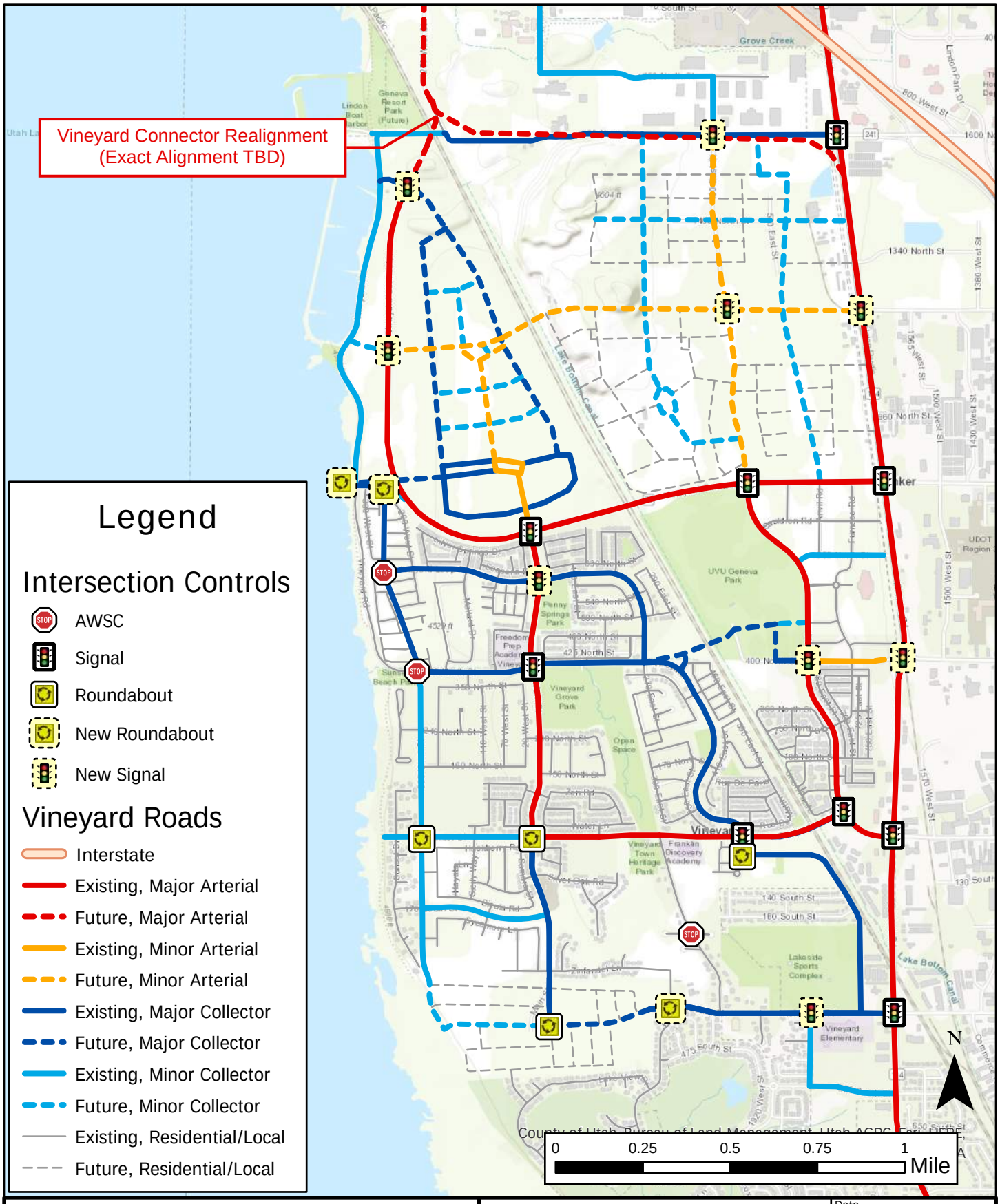
The permit may stipulate conditions and terms for the expiration of the permit when the necessity for the variance no longer exists, allowing direct access to a roadway when the access proposal cannot meet the standards of this access management manual, or when the property would otherwise be without reasonable alternative access if a variance were approved.

If a variance is denied, the applicant may initiate an appeal to the City Administrator, only after complete review and determination of the access permit application.

Appendix A

Vineyard City TMP

Future Roadway Network



VINEYARD
STAY CONNECTED
TRANSPORTATION MASTER PLAN

Future (2050) Roadway Network

Date 6/17/2025

Project UT23-2453

Appendix A